

Jiarong (Bill) Lu

226-989-5986 · j94lu@uwaterloo.ca

RELEVANT SKILL

Language: C, C++, VHDL, Lean, Python, HTML

Tools: VS code, GitHub, WordPress, Premiere Pro, Photoshop, DaVinci Resolve, Autodesk Inventor, OBS, Drones, After Effect, Logic Pro, Adobe Audition, CAD, Unreal Engine, VMIX, Ollama, Docker, Open WebUI, STM32CubeIDE, Dify

EXPERIENCE

Waterloo Aerial Robotics Group (WARG)

Ontario, Canada

Embedded Flight Software (EFS) develops team member

Oct 2024 - Present

- A student design team at the University of Waterloo designs drones or fixed-wing aerial robots.
- Succeed in building the motor driver of the new driver architecture. Now working on the Attitude Manager with PID control function, which is a module that smooths the pilot flying experience.

Shanghai Mindmatrix Technology Ltd

Shanghai, China

Embedded Flight Software (EFS) develops team member

Jun 2025 - Sept 2025

- Developed embedded firmware for BLDC motor control using STM32 microcontrollers, with real-time PWM, ADC, and communication via STM32CubeIDE and C/C++.
- Designed and fabricated custom multi-layer BLDC driver PCBs using Altium Designer; managed manufacturing and testing with JLCPCB.
- Built a mechanical torque testing rig using Autodesk Inventor to evaluate motor performance under controlled load conditions.
- Currently implementing advanced motor control algorithms, including Field-Oriented Control (FOC) and finely tuned PID loops, to achieve precise angular positioning with <0.1° error.
- Redesigning PCB layout and reselecting components to significantly reduce board size and meet integration requirements in constrained environments.

High School activity website

Shanghai, China

Founder and director

Sept 2021 - June 2024

- Built and maintained a school activity website supporting over 60 clubs using WordPress.
- Managing a team of six for weekly updates and maintenance.

Bopoda Technology Co., Ltd

Shanghai, China

Student Intern

July 2023 - Aug 2023

- Created 20 user guide videos for broadcasting platforms using Premiere Pro.
- Learned streaming technologies (OBS, VMIX) and the application of ChatGPT or relevant technologies in real-world business.

PROJECTS

Glory Media AI transformation

Remote | Jun 2025 - Aug 2025

- Collaborated in a 6-person team to develop a strategic AI transformation plan for Glory Media, focusing on automating content workflows in digital publishing
- Proposed the adoption of workflow tools such as Dify for automating tasks, including news summarization, article drafting, and editorial support
- Delivered a comprehensive final report outlining implementation steps, risks, and potential ROI, tailored to Glory Media's existing digital infrastructure
- Created a step-by-step Dify usage guide to facilitate internal adoption by non-technical editorial staff

Automatic Water Quality monitor

Ontario, Canada | Sept 2024 - Dec 2024

- Built a cost-effective water quality monitor using TDS sensors and STM32 microcontrollers
- Using C++ and C for coding the device and using Autodesk Inventor for building 3D model and printing the model

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals

Issued Jun 2025

Microsoft Certified: Azure Fundamentals

Issued Jun 2025

EDUCATION

Candidate for Bachelor of Applied Science, University of Waterloo

Sept 2024 - Expected: June 2029

Bachelor of Applied Science in Computer Engineering

Shanghai Weiyu High School

Sept 2021 - June 2024

IBDP (International Baccalaureate Diploma Programme)